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incident-Response Responsibilities

CampusGuard is responsible for coordinating the complete lifecycle of a campus emergency, from the initial reporting of an incident until its resolution. The system registers and tracks incidents, changes their operational status, activates or cancels emergency alerts, dispatches security, medical and facilities responders, coordinates communication and evacuation instructions, controls access to affected areas, and performs lockdown operations when required. It also supports specialised responder capabilities for hazardous environments and integrates incompatible legacy access-control hardware through a modern interface. These responsibilities allow the different campus emergency services to work together without becoming directly dependent on one another.

How the Six Design Patterns Interact

The six design patterns operate as parts of one emergency-response workflow rather than as separate demonstrations. The Facade provides the main entry point through CampusEmergencyFacade and coordinates the response. When a panic alert is received, it interacts with the Incident context, where the State pattern changes the incident from ReportedState to ActiveCrisisState and changes which operations, such as deployment and lockdown, are permitted. The Facade then uses the Command pattern through OperatorInvoker, which executes an appropriate command such as TriggerAlertCommand.

The command delegates the operational work to the Mediator, which coordinates the registered campus response components rather than allowing security, medical, facilities, communication and access-control components to communicate directly with one another. When responders such as SecurityTeam or MedicalResponder are deployed into specialised environments, the Decorator pattern adds capabilities such as low-visibility support without changing the underlying responder class. At the same time, access-control requests can pass through the Adapter, which translates the modern lock-control interface into commands understood by the legacy access hardware.

FAÇADE

CampusEmergencyFacade

STATE

- Incident-Context
- IncidentState- State
- ReportedState-concreteState1
- ActiveCrisisState-concreteState2
- Resolvedstate-ConcreteState3

COMMAND

- Command-EmergencyCommand
- Concrete Command:
- TriggerAlertCommand
- CancelAlertCommand
- LockdownZoneCommand
- Invoker:
- OperatorInvoker
- Receiver:
- EmergencyMediator

MEDIATOR

- Mediator-CampusMediator
- Concrete Mediator-EmergencyMediator
- Colleague-ResponseComponent
- ConcreteColleagues:
- VictimInterface
- SecurityTeam
- MedicalResponder
- FacilitiesStaff
- AccessControlSystem
- CommunicationService

ADAPTER

- Client-AccessControlSystem
- Target-ModernLockInterface
- Adapter-LegacySystemAdapter
- Adaptee-Old1998ServerHarware-

DECORATOR

- Component: EmergencyResponder
- Concrete Component: BaseResponderUnit
- Decorator: ResponderDecorator
- Concrete Decorators:
 - UnpoweredConstructionZoneDecorator
 - SirensEnabledDecorator
- Users of decorated components:
 - SecurityTeam
 - MedicalResponder

Ownership decision taken:

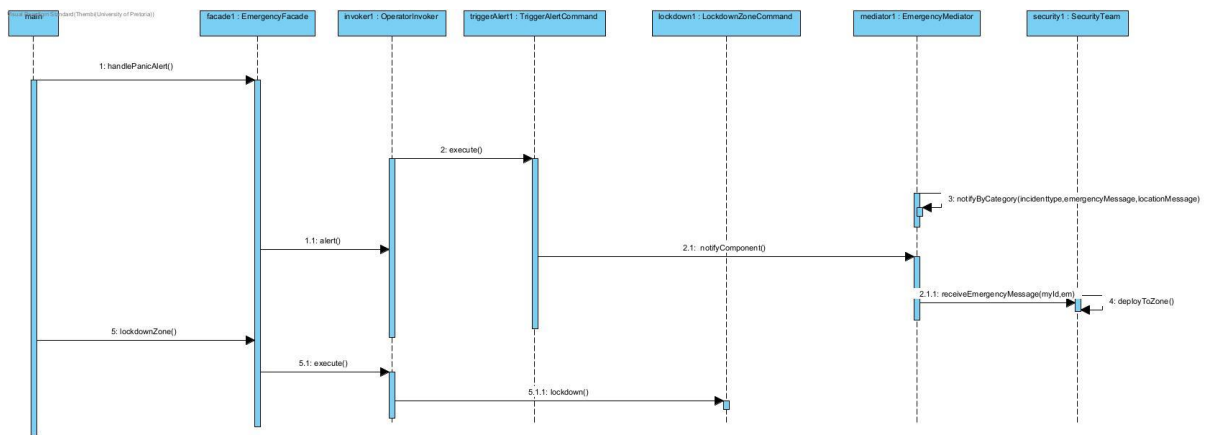
The code relies heavily on `std::unique_ptr` for building the **Decorator** and **Adapter** chains. This is a deliberate choice ensuring that only *one* entity owns a resource at a given time, preventing memory leaks when the chains are built.

Memory is managed safely by passing ownership up a chain

Task 4

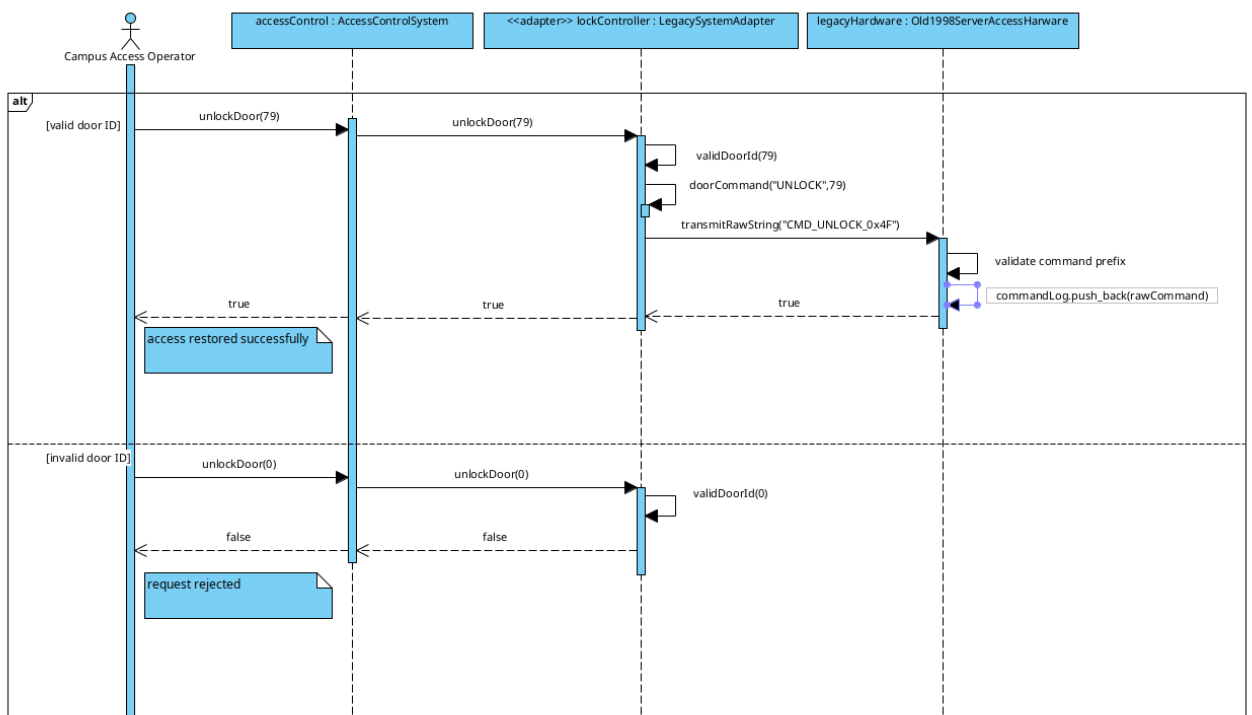
Paste the diagrams here

UML diagram



Sequence Diagram 2

sd [sequence diagram 2 : Restoring Access Through the Legacy Adapter]



DEBUGGING

```

Make breakpoint pending on future shared library load? (y or [n]) Y
Breakpoint 1 (tests/MainTest.cpp:181) pending.
(gdb) run
The program being debugged has been started already.
Start it from the beginning? (y or n) Y
Starting program: /app/bin/tests/main
warning: Error disabling address space randomization: Operation not permitted
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Adapter] Translating lockDoor(79) to CMD_LOCK_0x4F.
[Legacy Access] Transmitted raw command: CMD_LOCK_0x4F
[Adapter] Translating unlockDoor(79) to CMD_UNLOCK_0x4F.
[Legacy Access] Transmitted raw command: CMD_UNLOCK_0x4F
[Adapter] Translating restrictArea("construction_zone") to CMD_RESTRICT_CONSTRUCTION_ZONE.
[Legacy Access] Transmitted raw command: CMD_RESTRICT_CONSTRUCTION_ZONE
[Adapter] Cannot lock invalid door ID 0.
[Adapter] Cannot restrict invalid area code: invalid area.
access-control-1 received emergency: Security incident at Engineering Block
AccessControlSystem: restricting the incident perimeter and locking the nearest legacy-controlled door.
[Adapter] Translating restrictArea("EMERGENCY_PERIMETER") to CMD_RESTRICT_EMERGENCY_PERIMETER.
[Legacy Access] Transmitted raw command: CMD_RESTRICT_EMERGENCY_PERIMETER
[Adapter] Translating lockDoor(79) to CMD_LOCK_0x4F.
[Legacy Access] Transmitted raw command: CMD_LOCK_0x4F
AccessControlSystem: Legacy access operations acknowledged.
Campus lockdown initiated.
[Mediator] Coordinating evacuation: Use the eastern assembly point and assist injured students
security-team-1 received emergency from Engineering construction zone
SecurityTeam capabilities: standard emergency-response equipment, portable lighting, thermal tracking, construction-barrier override
emergency sirens, traffic-lane priority (clearance 4)
[Decorator] Emergency sirens enabled; traffic-lane priority requested.
[Decorator] Activating portable lighting and thermal tracking.
[Responder] Test security unit deploying to Engineering construction zone.
[Decorator] Construction-barrier override enabled for Engineering construction zone.

```

Use of docker and GDB

```

root@196fec03050b:/app# gdb bin/tests/main
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from bin/tests/main...
(No debugging symbols found in bin/tests/main)
(gdb) run
Starting program: /app/bin/tests/main
warning: Error disabling address space randomization: Operation not permitted
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Adapter] Translating lockDoor(79) to CMD_LOCK_0x4F.
[Legacy Access] Transmitted raw command: CMD_LOCK_0x4F
[Adapter] Translating unlockDoor(79) to CMD_UNLOCK_0x4F.
[Legacy Access] Transmitted raw command: CMD_UNLOCK_0x4F
[Adapter] Translating restrictArea("construction_zone") to CMD_RESTRICT_CONSTRUCTION_ZONE.
[Legacy Access] Transmitted raw command: CMD_RESTRICT_CONSTRUCTION_ZONE
[Adapter] Cannot lock invalid door ID 0.
[Adapter] Cannot restrict invalid area code: invalid area.

```

```

u25224663@LAPTOP-MKJ3RFD3:~/sem2/prac5/COS214-Practical5$ docker compose up --build
[+] Building 5.1s (11/11) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 558B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 192B
=> [internal] load metadata for docker.io/library/ubuntu:22.04
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 1.23MB
=> [1/4] FROM docker.io/library/ubuntu:22.04@sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6bb71b079fe399fde0329c02
=> => resolve docker.io/library/ubuntu:22.04@sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6bb71b079fe399fde0329c02
=> CACHED [2/4] RUN apt-get update && apt-get install -y g++ gdb make && rm -rf /var/lib/apt/lists/*
=> CACHED [3/4] WORKDIR /app
=> [4/4] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:60e87238381db8b250ed984946ff7a099ebc1458759f653afc247c7b5e86ee
=> => exporting config sha256:3275eba1546ea0ea763d5fa08cb7b05138aa76cd578dff7758d1085ebe776dc5
=> => exporting attestation manifest sha256:b063a834497b46217423ff1646a094bd406d1e938930bbb78f1a2b56a6063c61
=> => exporting manifest list sha256:c5b548fcaa733cb0770c43dc88ba5cdb3098ff40701cc0fa4111686d238bce4
=> => naming to docker.io/library/cpp-gdb:latest
=> => unpacking to docker.io/library/cpp-gdb:latest
=> resolving provenance for metadata file
[+] up 2/2
✓ Image cpp-gdb Built
✓ Container cos214-practical5-campusguard-1 Recreated
Attaching to campusguard-1
campusguard-1 | SCENARIO 1: Security threat in Engineering construction zone
campusguard-1 | Incident #101reported atEngineering construction zone
campusguard-1 | Facilities team responding to incident
campusguard-1 |
campusguard-1 | Incident #101

```